# DHRUV DUBEY

AI Engineer | Agentic AI, LLM Systems & RAG | College Park, MD
ddubey12@umd.edu | LinkedIn | GitHub | Portfolio

## EDUCATION

**University of Maryland, College Park** — M.S., Data Science (GPA 4.00/4.00) *Aug 2025 – May 2027*

**MIT-ADT University** — B.Tech, Computer Science & Engineering (GPA 3.33/4.00) *2021 – 2025*

## TECHNICAL SKILLS

**Languages:** Python, SQL, C++, R

**Agents & LLMs:** Agentic AI, Multi-Agent Systems, LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), Tool Calling, Prompt & Context Engineering, RAG, Vector Databases, Embeddings, LoRA/QLoRA Fine-Tuning, Open-Weight LLMs (Qwen3 via Ollama), Snowflake Cortex

**Evaluation:** LLM & Agent Evaluation, Golden Datasets, CI Eval Gates, Crash-Recovery Benchmarks, Regression Testing

**Backend & MLOps:** FastAPI, REST APIs, Pydantic, Microservices, Docker, Kubernetes (HPA, Probes), GitHub Actions CI/CD, MLflow, AWS, SLURM/HPC

**ML:** PyTorch, TensorFlow, Scikit-learn, Deep Learning, NLP, Computer Vision, Time Series, Anomaly Detection

**Data:** Snowflake, PostgreSQL, SQLite, Redis, MongoDB, Neo4j, Supabase, Power BI, Pandas, NumPy

## EXPERIENCE

**Allegis Global Solutions** — AI Automation Engineer Intern *Jun 2026 – Sep 2026*

- Built a multi-agent triage system on Snowflake Cortex (Agents, Search, Analyst) that traces pipeline failures through a dependency graph to the broken stored procedure; root-cause diagnosis ~65% faster (weeks to days).
- Engineered semantic views and a YAML instruction layer built from prior tickets so the agent handles new failure patterns without retraining.
- Deployed it into analysts' Power BI workflow as a chat interface, cutting fix-proposal time from hours to minutes.

**University of Maryland** — Research Intern, Neuro-Symbolic AI *Sep 2025 – Sep 2026*

- Co-developed a six-stage reproducibility audit of neuro-symbolic AI: screened 5,497 records to 1,304 studies and reproduced 85 codebases on the Zaratan HPC cluster (SLURM, Docker).
- Found inconsistencies in setups, hyperparameters and evaluation methods across reasoning frameworks; co-authored the paper, now under review at IEEE TNNLS.

**Fortytwo Labs** — Machine Learning Intern *Feb 2025 – Jul 2025*

- Built TensorFlow autoencoders on login behavior (+40% anomaly detection accuracy) and KMeans/DBSCAN clustering that surfaced 35% more outliers.
- Integrated the models with the Drools rule engine for automated risk scoring, cutting manual security reviews 33%.

**Intrainz** — Data Science Intern *May 2023 – Jul 2023*

- Built a crop recommendation system on 5,000+ records, raising decision-support precision 15% across 150 pilot farmers.

## PROJECTS

**Pipeline Incident Triage Graph** GitHub

*LangGraph, Model Context Protocol, FastAPI, Pydantic, Docker, Kubernetes, GitHub Actions, pytest*

- Open-source multi-agent triage: a tracer walks the pipeline DAG upstream from the symptom, then specialist and reviewer agents draft and check a fix behind a human-approval gate.
- CI gated on two 100-incident eval sets: root-cause accuracy 1.00; noisy logs drop category accuracy to 0.81, which the gate now tracks. Ships an MCP server and Kubernetes manifests.

**Agent Harness Runtime** GitHub

*Python, FastAPI, SQLite, Model Context Protocol, Pydantic, Docker, Kubernetes, GitHub Actions*

- Agent runtime that owns the loop, context budget and per-agent tool sandbox, and logs every tool call so a crashed run resumes from its log instead of restarting. Validated with Qwen3 8B on Ollama.
- 200-run crash benchmark: 0 duplicated side effects vs 145 for restart-from-scratch. Ran the triage agent inside it: across 75 approved fixes killed mid-apply, 0 duplicated actions vs 61 on restart; 30 paused for human review.

**Krishi Mitra — Multilingual Agricultural RAG Assistant** GitHub

- Fine-tuned LLaMA 3.2 3B with RAG, answering farmers in Hindi, Marathi and English; piloted with 300 farmers, +20% query-resolution accuracy.

**Smart Plant Health Monitor** GitHub

- YOLOv5, ResNet and LSTM system classifying plant health at 90% accuracy, served as a Dockerized FastAPI microservice.

## PUBLICATIONS

- Neuro-Symbolic AI Reproducibility Review — University of Maryland; arXiv:2608.26236; under review, IEEE Transactions on Neural Networks and Learning Systems.
- Krishi Mitra: LLM-Driven QA System for Indian Farmers — MIT-ADT University.